

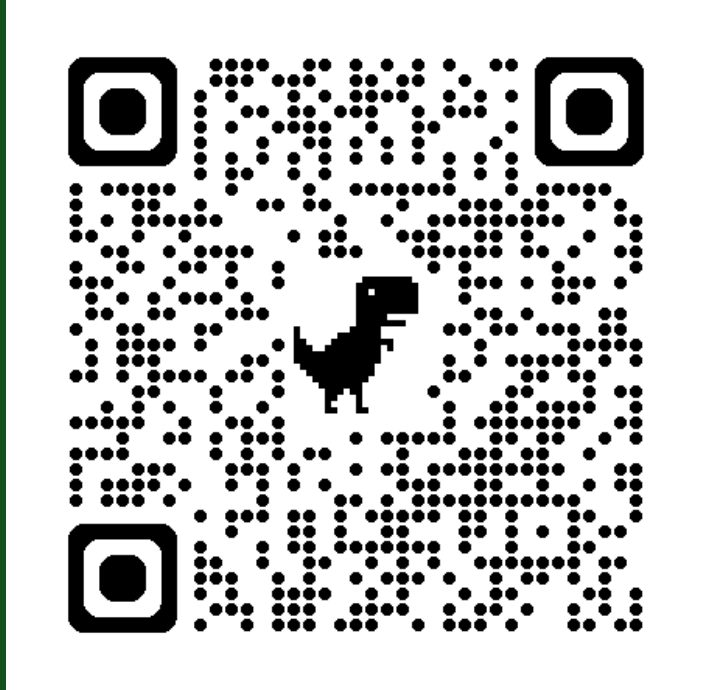
Spatial Clustering of Traumatic Experiences and Posttraumatic Stress Symptoms

UCLA

Evidence from Hurricane-Exposed Adults in Florida

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Background

Hurricanes produce geographic hotspots of physical destruction and human exposure. Whether these spatial patterns extend to posttraumatic stress symptoms (PTS) remains unclear.

Objective

Examine whether hurricane exposure and PTS symptoms show spatial clustering across Florida.

Method

Participants and procedure

- Probability-based sample of 1,052 Florida adults from Ipsos KnowledgePanel, data collected April 30–May 19, 2025.

Measures

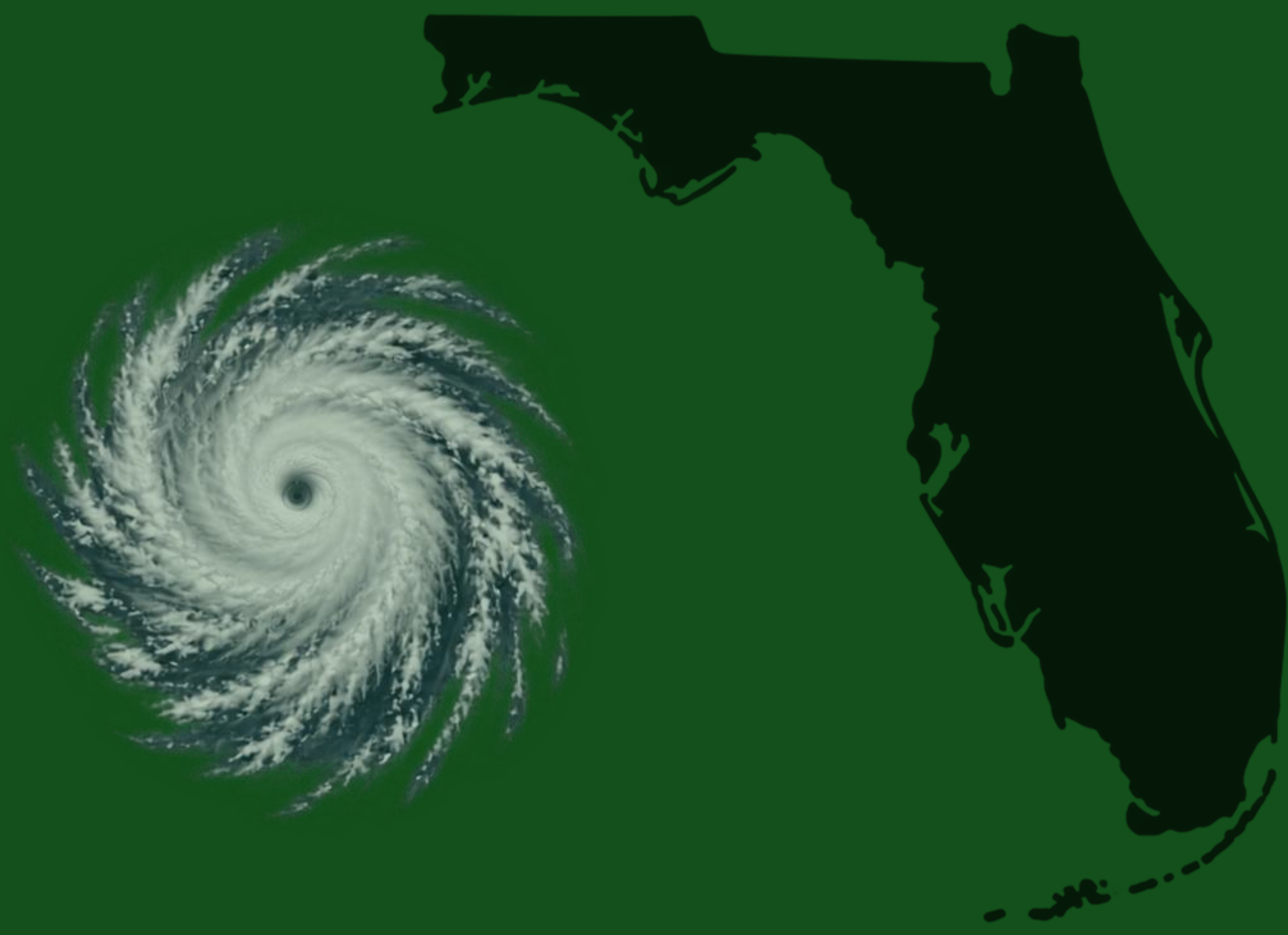
- **Hurricane exposure:** 10 yes/no prior hurricane experiences (e.g., evacuation, property damage, injury) summed into a cumulative exposure score.
- **PTS symptoms:** Five hurricane-related symptoms rated 0 (*never*)–4 (*all the time*) and averaged (modified PC-PTSD-5)

Spatial Analysis

- Spatial neighbors defined using 8-nearest neighbors
- Global Moran's I assessed spatial autocorrelation
- LISA (Local Indicators of Spatial Association) identified local hurricane-exposure clusters

Hurricane exposure
clusters geographically.

PTS symptoms do not.



Hurricane Exposure

Global Moran's I = 0.11

$p < .001$

Significant positive spatial
autocorrelation

PTS symptoms

Global Moran's I = -0.001

$p = .51$

No evidence of spatial
autocorrelation

Results

Descriptive Statistics

- Participants were 18–99 years old ($M = 62.7$, $SD = 14.36$).
- 51.0% bachelor's degree or above, 32.3% some college, 14.7% high school, and 1.9% less than high school.
- 59.1% were female and 40.9% male.
- 80.8% were White, 10.2% Hispanic, 6.4% Black, 1.4% multiracial, and 1.2% other race.

Hurricane exposure was spatially clustered, Global Moran's I = .11, $p < .001$. Participants with similar levels of hurricane exposure tended to live near one another.

PTS symptoms were not spatially clustered, Moran's I = -.001, $p = .51$. Participants with similar levels of PTS symptoms did not tend to live near one another.

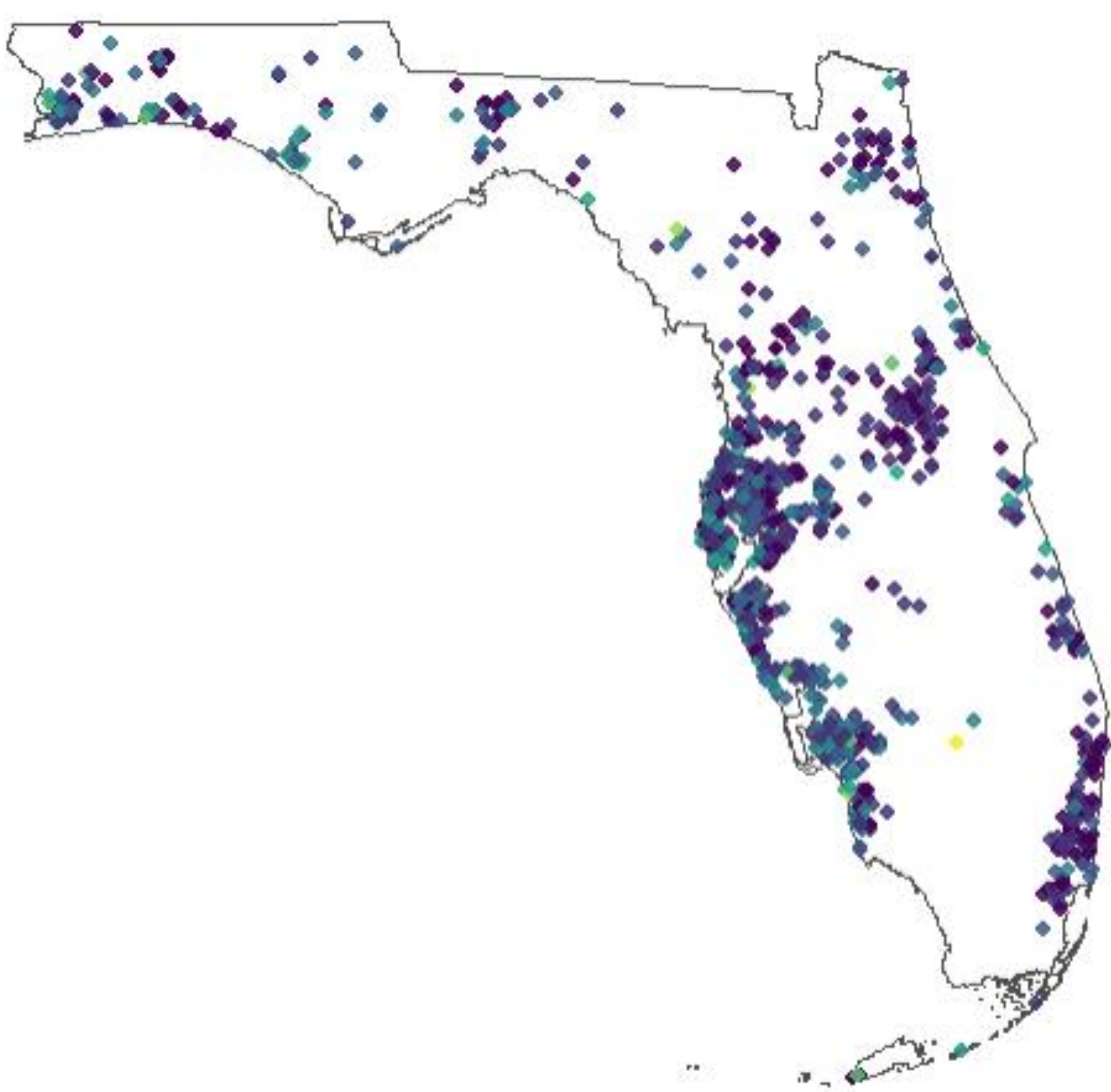
Greater hurricane exposure was associated with greater PTS symptoms, $r = .24$, 95% CI [.19, .30], $p < .001$. Thus, exposure and PTS were associated at the individual level, despite showing different patterns of geographic clustering.

Take-home message

Hurricane exposure forms geographic hotspots, but PTS symptoms do not. This highlights the need for geographically broad mental health screening beyond areas of greatest hurricane exposure.

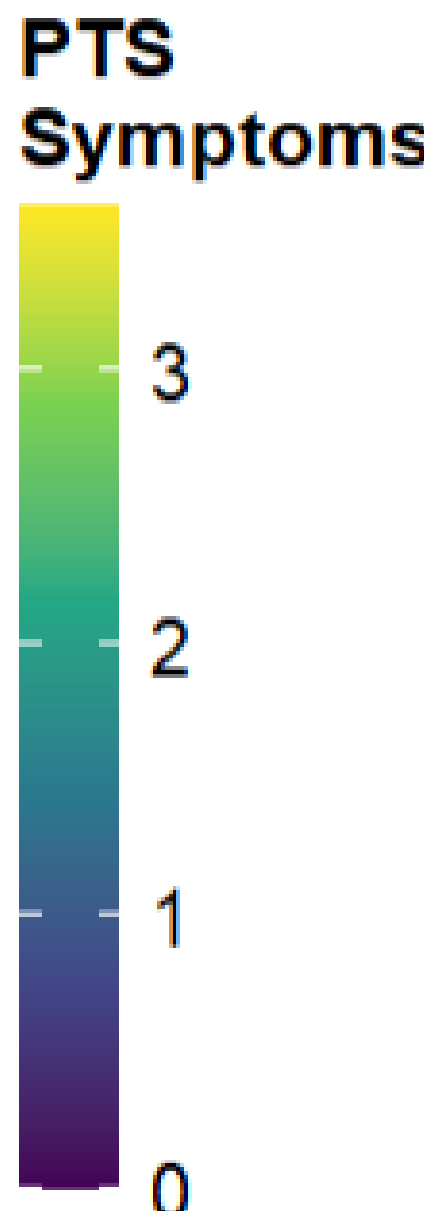
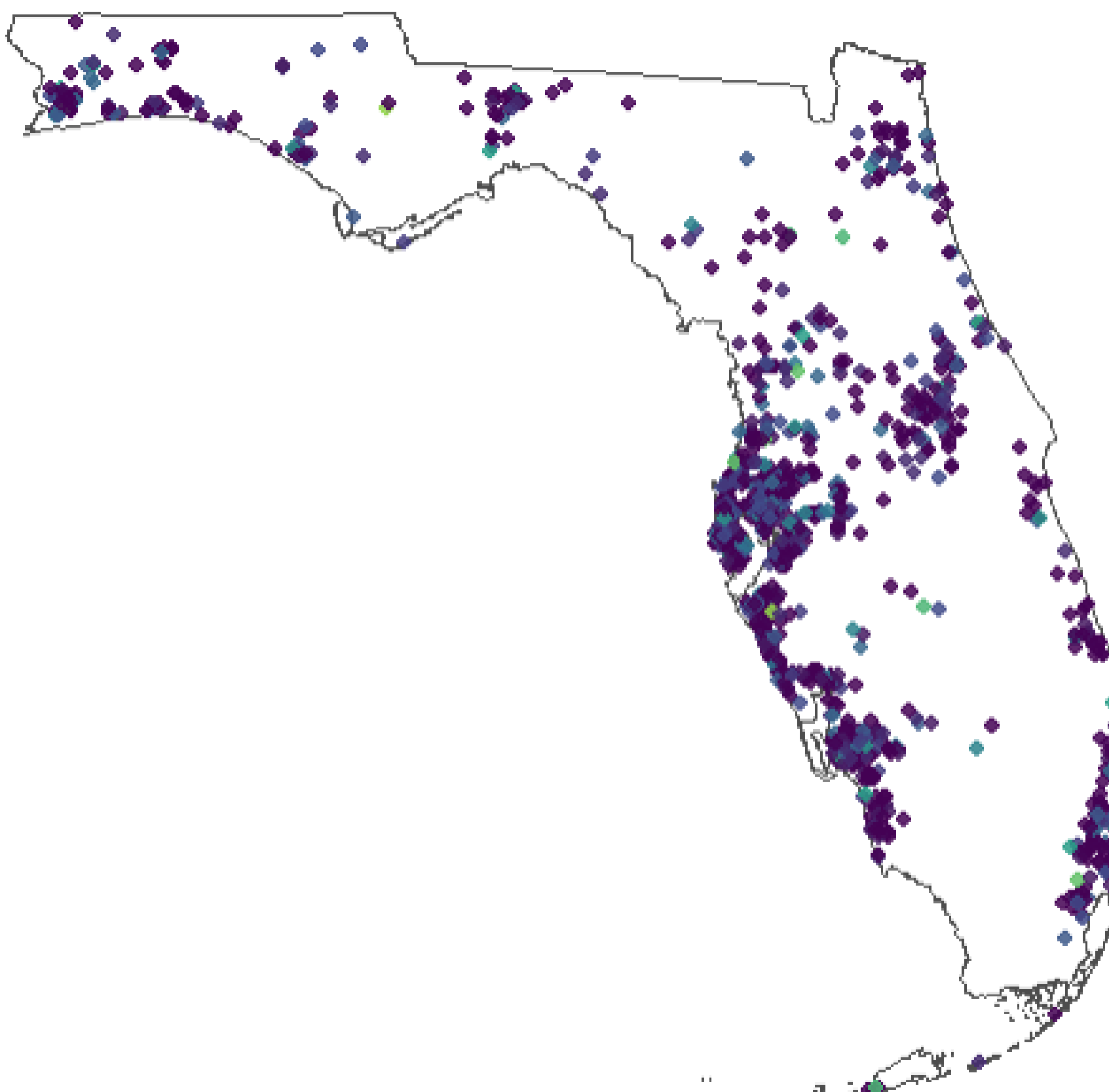
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Hurricane Exposure



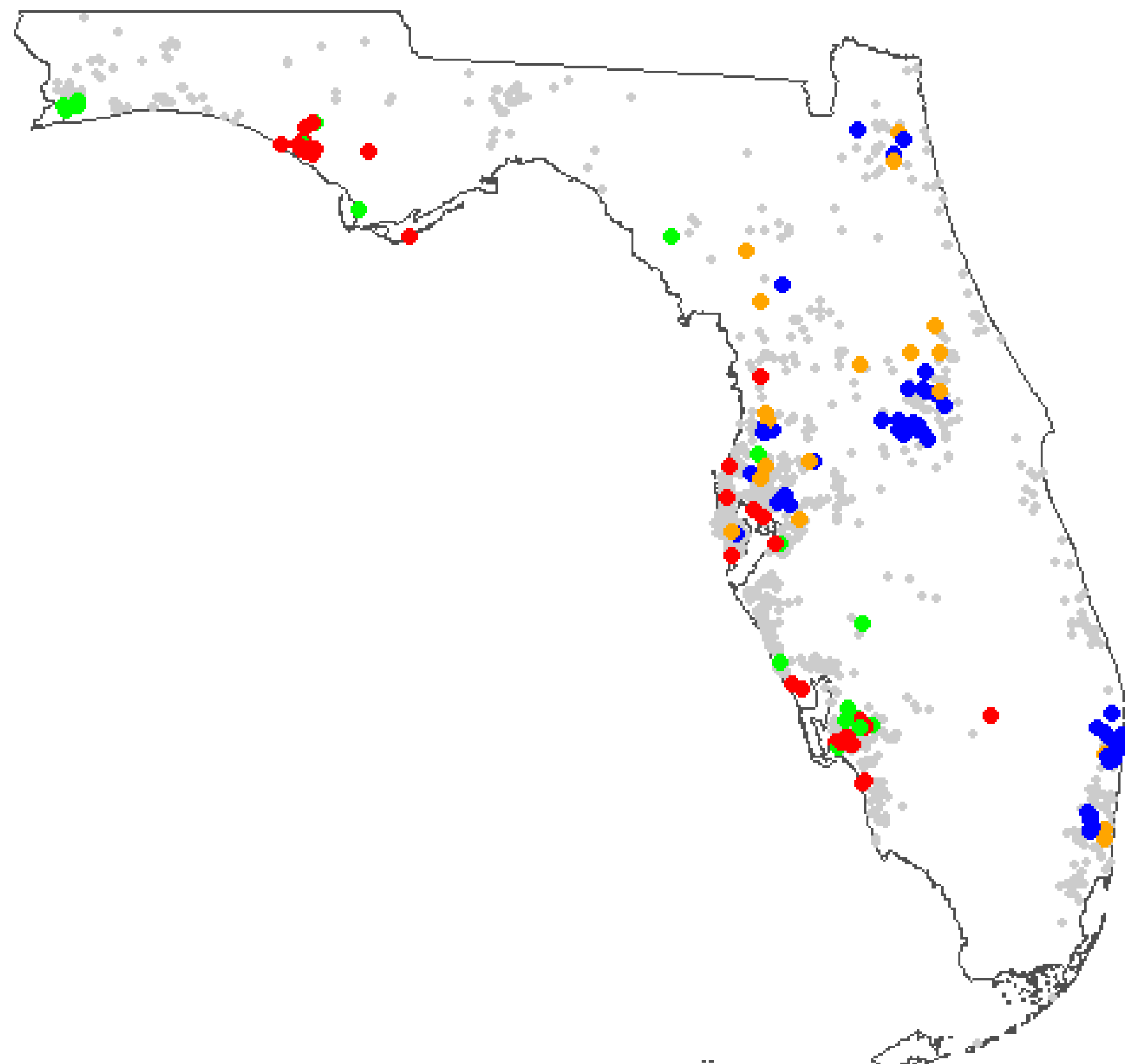
Each point represents a participant.
Color indicates cumulative hurricane exposure (0–7).

PTS Symptoms



Each point represents a participant.
Color indicates PTS symptom severity.

Hurricane Exposure Clusters



LISA Cluster

- High-High
- High-Low
- Low-High
- Low-Low

High-exposure participants surrounded by other high-exposure participants
Low-exposure participants surrounded by other low-exposure participants